

# SONGYUAN ZHANG

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## RESEARCH INTERESTS

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Reinforcement Learning (Online/Offline), Robot Safety, Learning-based Control, Generative Policies, Embodied AI, Humanoid Whole-Body Control, Imitation Learning, Multi-Agent Systems, Control Theory

## EDUCATION

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|------------------------------------------------------------------|----------------------------------|
| <b>Massachusetts Institute of Technology</b>                     | Sep. 2021 - May 2027 (expected)  |
| <i>Ph.D. Student, Department of Aeronautics and Astronautics</i> | <i>Advisor: Prof. Chuchu Fan</i> |

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|----------------------------------------------------------|----------------------------------|
| <b>Massachusetts Institute of Technology</b>             | Aug. 2021 - Feb. 2024            |
| <i>Master of Science in Aeronautics and Astronautics</i> | <i>Advisor: Prof. Chuchu Fan</i> |

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| <b>Tsinghua University</b>                                             | Sep. 2017 - Jul. 2021 |
| <i>B.E. in Tsien Excellence in Engineering Program (honors degree)</i> |                       |
| <i>Advisors: Prof. Yanan Sui, Prof. Hao Wu, Prof. Gangtie Zheng</i>    |                       |

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| <b>Stanford University</b>                         | Jun. 2020 - Jul. 2021              |
| <i>Remote Visiting Scholar in Computer Science</i> | <i>Advisor: Prof. Dorsa Sadigh</i> |

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| <b>University of California, Berkeley</b>         | Jul. 2019 - Sep. 2019                    |
| <i>Visiting Scholar in Mechanical Engineering</i> | <i>Advisor: Prof. Masayoshi Tomizuka</i> |

## RESEARCH AND PROFESSIONAL EXPERIENCE

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| <b>Amazon Industrial Robotics Group</b> | Summer 2026                                      |
| <i>Applied Scientist II Intern</i>      | <i>Project: Humanoid Whole-Body Safe Control</i> |

- Developed FAITH (Feasibility-Aware Safety-Filtered RL for High-Dimensional Systems), a model-free RL framework that separates learned safety filtering from task-policy optimization, allowing the policy to maximize task performance without a competing safety penalty.
- Deployed FAITH on a 29-DoF Unitree G1 humanoid with onboard sensing. Demonstrated walking, push recovery, and falls away from protected regions using a single controller without explicit mode switching.

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| <b>Massachusetts Institute of Technology</b> | Sep. 2021 - Now                                                      |
| <i>Doctoral Research</i>                     | <i>Thesis: Structured Policy Learning for Reliable Robot Control</i> |

- Formulated compositional stability certificates (NeurISS) and graph-based safety certificates (GCBF), enabling scalable, distributed control of large-scale multi-agent systems.
- Designed safe multi-agent control and RL algorithms (GCBF+, DGPPPO, Def-MARL), scaling to 1,024 agents in simulation and demonstrating collision avoidance and collaborative tasks on Crazyflie drones.
- Developed support-constrained offline RL algorithms (ReFORM, LASER) that optimize flow-based generative policies in latent space, preserving learned action support to mitigate out-of-distribution errors in locomotion and manipulation tasks.

## PUBLICATIONS

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\* *Equal contribution.*

[**Preprint**] Songyuan Zhang, Baljeet Singh, Sarthak Ranjeet Kaingade, Chuchu Fan, and Bryan Trinh. “FAITH: Feasibility-Aware Safety-Filtered RL for High-Dimensional Systems.” *arXiv preprint arXiv:2610.12432*, 2026. [[arXiv](#)]

[**NeurIPS’26**] Songyuan Zhang, Oswin So, Eric Yang Yu, Matthew Cleaveland, Peter Crowley-Dolen, and Chuchu Fan. “LASER: Latent Space Adjoint Matching for Support-Constrained Entropy-Regularized Offline RL.” *Conference on Neural Information Processing Systems (NeurIPS)*, 2026. [[Website](#)]

[**CDC’26**] Hao Wang, Max Cohen, Zeyuan Feng, Donggeon David Oh\*, Oswin So\*, Songyuan Zhang\*, Javier Borquez, Chuchu Fan, Jaime Fernández Fisac, Haimin Hu, Jason J. Choi, and Somil Bansal. “Handbook on Safety Certificates: Techniques from Hamilton–Jacobi Reachability Analysis and Control Barrier Functions.” *65th IEEE Conference on Decision and Control (CDC)*, 2026.

[**RA-L’26**] Jaeyoun Choi, Oswin So, Songyuan Zhang, Cooper Taylor, and Chuchu Fan. “Safe and Scalable Multi-Drone Payload Transport via CBF-based Reinforcement Learning with Zero-Shot Sim-to-Real Transfer.” *IEEE Robotics and Automation Letters (RA-L)*, 2026. [[PDF](#)]

[**TIA’26**] Qing Shen, Peng Zhang, Songyuan Zhang, Chuchu Fan, Yifan Zhou, and Yacov A. Shamash. “Hamiltonian Grid-Forming: Methodology and Applications for IBR-Rich Grid Operations.” *IEEE Transactions on Industry Applications*, 2026. [[PDF](#)]

[**ICRA’26**] Akila Saravanan, Songyuan Zhang, Travis Manderson, Nicholas Roy, and Chuchu Fan. “Beyond Waypoints: Semantic-Centric Autonomy with Unreliable Maps Through Learned Abstractions.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. [[PDF](#)]

[**ICLR’26**] Songyuan Zhang, Oswin So, H. M. Sabbir Ahmad, Eric Yang Yu, Matthew Cleaveland, Mitchell Black, and Chuchu Fan. “ReFORM: Reflected Flows for On-support Offline RL via Noise Manipulation.” *International Conference on Learning Representations (ICLR)*, 2026. [[Website](#)]

[**ICLR’26**] Oswin So, Eric Yang Yu, Songyuan Zhang, Matthew Cleaveland, Mitchell Black, and Chuchu Fan. “Solving Parameter-Robust Avoid Problems with Unknown Feasibility using Reinforcement Learning.” *International Conference on Learning Representations (ICLR)*, 2026. [[Website](#)]

[**NeurIPS’25**] H. M. Sabbir Ahmad, Ehsan Sabouni, Alexander Wasilkoff, Param Budhraj, Zijian Guo, Songyuan Zhang, Chuchu Fan, Christos Cassandras, and Wenchao Li. “HMARL-CBF – Hierarchical Multi-Agent Reinforcement Learning with Control Barrier Functions for Safety-Critical Autonomous Systems.” *The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025. [[Website](#)]

[**RSS’25 (Outstanding Student Paper Award)**] Songyuan Zhang\*, Oswin So\*, Mitchell Black, Zachary Serlin, and Chuchu Fan. “Solving Multi-Agent Safe Optimal Control with Distributed Epigraph Form MARL.” *Proceedings of Robotics: Science and Systems (RSS)*, 2025. [[Website](#)]

[**ICLR’25**] Songyuan Zhang, Oswin So, Mitchell Black, and Chuchu Fan. “Discrete GCBF Proximal Policy Optimization for Multi-agent Safe Optimal Control.” *International Conference on Learning Representations (ICLR)*, 2025. [[Website](#)]

[**T-RO’25**] Songyuan Zhang\*, Oswin So\*, Kunal Garg, and Chuchu Fan. “GCBF+: A Neural Graph Control Barrier Function Framework for Distributed Safe Multi-Agent Control.” *IEEE Transactions on Robotics*, 2025. [[Website](#)]

[**L4DC’24**] Songyuan Zhang and Chuchu Fan. “Learning to stabilize high-dimensional unknown systems using Lyapunov-guided exploration.” *6th Annual Learning for Dynamics & Control Conference (L4DC)*, 2024. [[Website](#)]

- [**ARC'24**] Kunal Garg, Songyuan Zhang, Oswin So, Charles Dawson, and Chuchu Fan. “Learning Safe Control for Multi-Robot Systems: Methods, Verification, and Open Challenges.” *Annual Reviews in Control*, 2024. [[arXiv](#)]
- [**CoRL'23**] Songyuan Zhang, Kunal Garg, and Chuchu Fan. “Neural Graph Control Barrier Functions Guided Distributed Collision-avoidance Multi-agent Control.” *7th Conference on Robot Learning (CoRL)*, 2023. [[Website](#)]
- [**SG'23**] Lizhi Wang, Songyuan Zhang, Yifan Zhou, Chuchu Fan, Peng Zhang, and Yacov A. Shamash. “Physics-Informed, Safety and Stability Certified Neural Control for Uncertain Networked Microgrids.” *IEEE Transactions on Smart Grid*, 2023. [[PDF](#)]
- [**PESGM'23**] Lizhi Wang, Songyuan Zhang, Yifan Zhou, Chuchu Fan, Peng Zhang, and Yacov A. Shamash. “Learning-Based, Safety and Stability-Certified Microgrid Control.” *2023 IEEE Power & Energy Society General Meeting (PESGM)*, 2023. [[PDF](#)]
- [**L4DC'23 (ORAL)**] Songyuan Zhang, Yumeng Xiu, Guannan Qu, and Chuchu Fan. “Compositional Neural Certificates for Networked Dynamical Systems.” *5th Annual Learning for Dynamics & Control Conference (L4DC)*, 2023. [[Website](#)]
- [**NeurIPS'21**] Songyuan Zhang\*, Zhangjie Cao\*, Dorsa Sadigh, and Yanan Sui. “Confidence-Aware Imitation Learning from Demonstrations with Varying Optimality.” *Conference on Neural Information Processing Systems*, 2021. [[Website](#)]

## HONORS AND AWARDS

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|------------------------------------------------------------------------------------------|------|
| 2025-2026 Schwarzman College of Computing Amazon AI Research Innovation Fellowship       | 2025 |
| Outstanding Student Paper Award, Robotics: Science and Systems (RSS)                     | 2025 |
| Outstanding Paper Award Finalist, Robotics: Science and Systems (RSS)                    | 2025 |
| Outstanding Graduate of Tsinghua University                                              | 2021 |
| Outstanding Graduate of Beijing                                                          | 2021 |
| Outstanding Graduate of Scientific Research at TEEP                                      | 2021 |
| Outstanding Graduate of Creativity at TEEP                                               | 2021 |
| Honor Degree of Tsien Class, Tsinghua University                                         | 2021 |
| Tsinghua Scholarship for Technological Innovation                                        | 2020 |
| National Scholarship, China                                                              | 2019 |
| First Prize in Excellent Student Research Training Contest of Tsinghua University        | 2019 |
| First Prize in "Challenge Cup" Science and Technology Competition of Tsinghua University | 2019 |

## ACADEMIC & PROFESSIONAL SERVICE

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**Workshop Organizing Committee Member:** [Workshop on Safety Analysis for Autonomous System: Practical Perspectives and Recent Advances](#), 65th IEEE Conference on Decision and Control (CDC), 2026

**Panel Moderator:** [Robotics × Physical AI Founder-Investor Summit](#) – Academia panel on frontier research, world models, and safe autonomy (July 2026)

### Journal Reviewer:

IEEE Transactions on Robotics (T-RO), IEEE Transactions on Automatic Control (T-AC), Automatica, Annual Reviews in Control (ARC), IEEE Open Journal of Control Systems (OJ-CSYS), IEEE Robotics and Automation Letters (RA-L), IEEE Transactions on Cybernetics (T-CYB), IEEE Transactions on Control Systems Technology (TCST)

**Conference Reviewer:**

NeurIPS, ICLR, ICML, RSS, CoRL, ICRA, L4DC, IROS, AAAI, AISTATS, CDC, ACC, NeuS, NFM, DARS, RV, TACAS, ANTS, CCTA

**Teaching Assistant:** X-idea, Tsinghua University (Spring 2021)

**LEADERSHIP & MEMBERSHIP**

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Chairman of 13th *Spark* Innovative Talents Training Program at Tsinghua University

Vice-monitor of Tsien Class 2017

Member of the Second Team of Middle and Long Distance Running at Tsinghua University

**SELECTED PRESS COVERAGE**

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MIT engineers help multirobot systems stay in the safety zone, *MIT News*, 2025. [\[Website\]](#)

Nine MIT PhD students named Amazon AI Fellows, *MIT Schwarzman College of Computing*, 2025. [\[Website\]](#)

**HOBBIES**

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Travelling, Hiking, Photography, Badminton, Running, Skiing